

Agentic AI Translation Company 2.0 — Production Pipeline

POST /api/translate → run_job() · events streamed to the client as NDJSON · every LLM call: OpenRouter / OpenAI / Anthropic / Gemini / xAI, up to 3 retries

1 · Order intake (backend/main.py)

Term base (optional)

CSV · TSV · XLSX · JSON · TXT
Parsed to {source term → target term}

Style sheet (optional)

TXT · MD · DOCX · PDF · HTML · JSON
Extracted to plain text (≤ 6,000 chars in prompts)

Client order

Source document: PDF · DOCX · PPTX · HTML · JSON · MD · TXT
Source → target language · translation type · brief
Output format · provider · model · API key
Chunk size · per-specialist temperatures · prompt overrides

Validate & store uploads

File-type checks · per-job folder (24 h TTL)

2 · Preparation (app/files.py)

Text extraction

Document → plain text

Chunking

Recursive split: paragraph → line → sentence → word
Chunk size clamped 300–12,000 chars · no overlap

passage 1 sample

3 · Project analysis (RealBrain.analyze)

Project Manager

LLM · temperature 0.4
Reads the brief + first 1,800 chars of passage 1
Knows whether a term base / style sheet was supplied

Project profile (JSON)

domain · detected style · target reader · locale
style guidelines (3–5) · naturalness pitfalls (2–4)
quality requirements (2–4) · reasoning · client reply

Client chose a translation type?

yes

no (Help me to decide)

Use client's type

e.g. Legal → legal

Use manager's domain

normalised; fallback → master

Route to one specialist team

team_for_domain(domain)

guidelines · requirements · profile

relevant terms per passage

4 · Translation (passage queue, team members work in parallel)

News & Media

News (0.7)
2 translators

Creative Studio

Literary (0.8) · Marketing (0.8)
2 translators

Legal & Business

Legal (0.65) · Business (0.7)
2 translators

Science & Medical

Academic (0.7) · Technical (0.65) · Medical (0.6)
2 translators

Master Translator

Mixed / unusual content (0.7)
1 translator

Translator prompt (per passage)

Role prompt (editable) + naturalness standard
+ language-pair notes (EN→zh-TW / zh-HK, zh→EN)
+ domain rules + manager guidelines + reader profile
+ term-base entries found in this passage
+ style sheet + end of previous passage (context)

every passage

5 · Editing (editor queue · 2 parallel slots)

Senior Editor · Review

LLM · temperature 0.3
Pass 1 — native read: translationese, transliteration, locale
Pass 2 — fidelity vs. source, term base, style sheet
Output: tagged issues with suggested rewrites

review notes

Senior Editor · Improve

LLM · temperature 0.4
Applies review notes · may rewrite whole sentences
Keeps meaning, term base and style sheet

6 · Terminology (term queue · 2 parallel slots)

Term base uploaded?

yes

Terms appear in passage?

yes

no

Terminologist

LLM · temperature 0.1
Swaps non-compliant terms for term-base entries
Adjusts only neighbouring words for grammar

Pass through unchanged

Edited passage is final

no

7 · Binding & delivery

Binding

Waits for every passage · reassembles in source order

Translation file

DOCX · TXT · MD · HTML · JSON

Job details report (JSON)

Analysis · team · per-passage steps (draft → review → revision → terms)
Token usage · duration

Client download

GET /api/download/{job_id}/translation | details